Data Displays

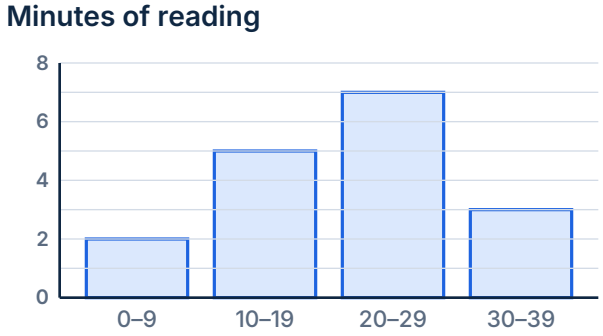
Ask a question with many answers. Then show the data.

A statistics question expects different answers

“How tall are the kids in my class?” Yes: many answers.

“How tall is Mia?” No: one answer.

A histogram: count the data in equal groups



Each bar is a group of numbers.

7 kids read 20 to 29 minutes.

The bars touch: the groups follow on.

One far-off number pulls the mean, not the median

2, 3, 3, 4, 28: mean 8, median 3

Your turn!

1, 4, 4, 6, 15: mean = 6 and median = 4

For teachers

Goal: statistics questions, X charts and histograms, and how a far-off value moves the mean (6.SP.1, 6.SP.4, 6.SP.5). Say “Does this question have one answer or many?”

Data Displays: Practice

Group the data. Show it. Read it.

Part 1 Statistics question? Many different answers? Circle yes or no.

1)	How far can sixth graders jump?	**yes** / no
2)	How many hours of sleep do kids in my class get?	**yes** / no
3)	How tall is Mia?	yes / **no**
4)	How tall are the trees in the park?	**yes** / no
5)	What time does school start today?	yes / **no**
6)	How many days are in a week?	yes / **no**
7)	How many pets do families on my street have?	**yes** / no
8)	How many legs does a spider have?	yes / **no**

Data Displays: Practice (continued)

Part 2 Count the groups How many numbers fall in each group?

9) **12, 25, 7, 18, 31, 22, 9, 15, 27, 3, 19, 24**

0–9 [3] 10–19 [4] 20–29 [4] 30–39 [1]

10) **45, 52, 61, 48, 57, 66, 50, 43, 59, 63**

40–49 [3] 50–59 [4] 60–69 [3]

11) **4, 11, 8, 15, 2, 13, 6, 9, 17, 12, 5, 10**

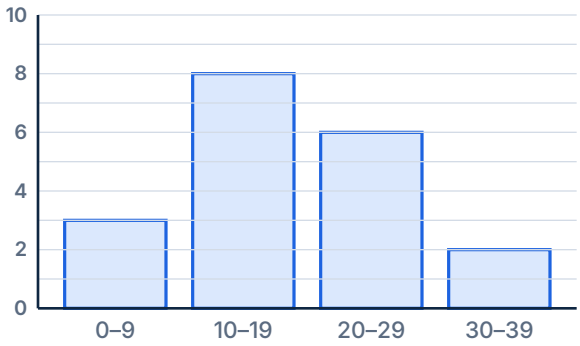
0–4 [2] 5–9 [4] 10–14 [4] 15–19 [2]

12) **33, 21, 38, 26, 29, 35, 24, 31, 37, 22**

20–24 [3] 25–29 [2] 30–34 [2] 35–39 [3]

Part 3 Read the histogram Each bar is a group.

13) **Minutes on the bus**



How many in 20–29? [6]

Most in which group? [10–19]

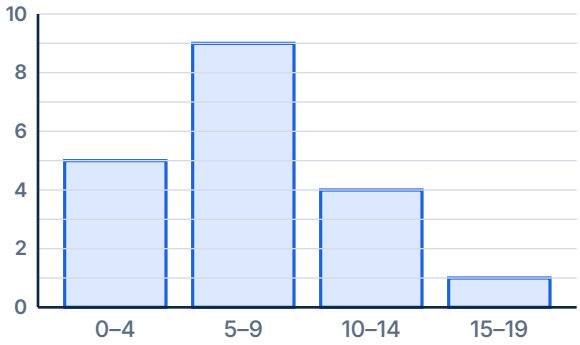
How many are 20 or more? [8]

How many in all? [19]

Data Displays: Practice (continued)

Part 3 Read the histogram (continued)

14) **Points in a game**



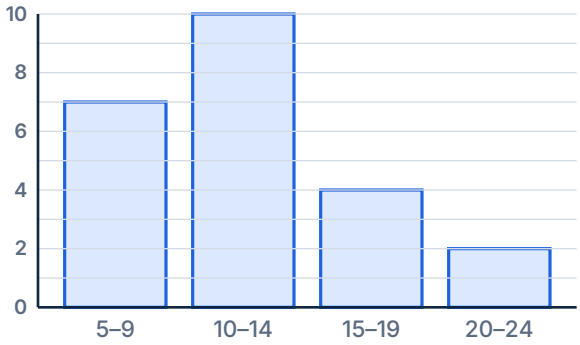
How many in 10–14? **4**

Most in which group? **5–9**

How many are 10 or more? **5**

How many in all? **19**

15) **Ages at the center**



How many in 15–19? **4**

Most in which group? **10–14**

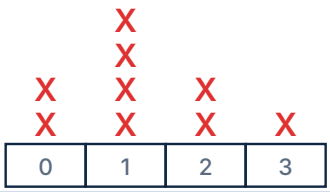
How many are 15 or more? **6**

How many in all? **23**

Part 4 Make the X chart One X for each number.

16) **2, 0, 1, 3, 1, 1, 2, 0, 1**

Siblings



mode = **1**

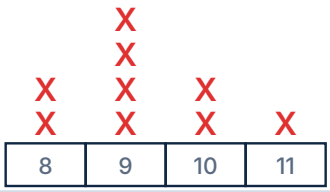
range = **3**

Data Displays: Practice (continued)

Part 4 Make the X chart (continued)

17) **9, 8, 10, 9, 9, 8, 11, 10, 9**

Hours of sleep

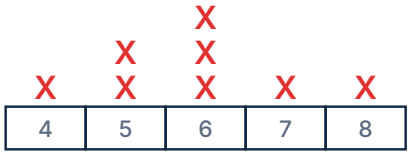


mode = 9

range = 3

18) **5, 6, 4, 6, 7, 5, 6, 8**

Shoe sizes



mode = 6

range = 4

Part 5 A far-off number Find both. Which one moved?

19) **3, 4, 4, 5, 24: mean = 8 and median = 4**

20) **10, 12, 12, 14, 52: mean = 20 and median = 12**

21) **1, 2, 2, 3, 17: mean = 5 and median = 2**

22) **6, 7, 8, 8, 31: mean = 12 and median = 8**

23) **20, 21, 21, 22, 66: mean = 30 and median = 21**

Data Displays: Practice (continued)

Part 6 More far-off numbers

24) 34, 35, 37, 36, 68: mean = 42 and median = 36

25) 14, 92, 13, 12, 14: mean = 29 and median = 14

26) 36, 34, 31, 31, 78: mean = 42 and median = 34

27) 22, 23, 19, 46, 20: mean = 26 and median = 22

28) 24, 29, 25, 97, 25: mean = 40 and median = 25

29) 21, 25, 95, 21, 23: mean = 37 and median = 23

30) 37, 55, 36, 37, 40: mean = 41 and median = 37

31) 9, 6, 8, 6, 76: mean = 21 and median = 8

32) 47, 10, 7, 5, 6: mean = 15 and median = 7

33) 8, 82, 11, 10, 9: mean = 24 and median = 10

Data Displays: Practice (continued)

Part 7 More means and medians

34) mean of 41, 53, 20, 46 = 40

35) mean of 39, 15, 56, 26 = 34

36) mean of 25, 35, 50, 42 = 38

37) mean of 13, 28, 44, 39 = 31

38) mean of 34, 22, 60, 20 = 34

39) mean of 55, 23, 14, 40 = 33

40) mean of 56, 46, 45, 17 = 41

41) mean of 58, 47, 13, 50 = 42

42) mean of 22, 46, 59, 29 = 39

43) mean of 13, 33, 35, 15 = 24

44) mean of 32, 10, 17, 49 = 27

45) mean of 37, 44, 24, 27 = 33

Data Displays: Practice (continued)

Part 8 Remember? Problems from earlier packets.

46) mean of 8, 13, 9, 14 = **11**

47) $6 \times 5 \times 2 + 2 \times 4 =$ **68**

48) $3\frac{1}{2} \times 2 \times 4 =$ **28**

49) $x + 47 = 92$: $x =$ **45**

50) $3{,}456 \div 12 =$ **288**

51) -15 **<** 2

Part 9 Word problems Which measure? Then find it.

52) Test scores: 82, 90, 75, 90, 88. What is the median?

Number sentence **75, 82, 88, 90, 90**

Answer **88** points

Draw or show your work

53) Kids read 3, 5, 4, 5, 8 books. What is the mean?

Number sentence **3 + 5 + 4 + 5 + 8 = 25, 25 ÷ 5 = 5**

Answer **5** books

Draw or show your work

54) In a histogram the bars show 4, 9, 6 and 1 kids. How many kids in all?

Number sentence **4 + 9 + 6 + 1 = 20**

Answer **20** kids

Draw or show your work

Data Displays: Practice (continued)

Part 9 Word problems (continued)

55) Heights: 50, 52, 52, 55, 56 in. What is the range?

Number sentence 56 − 50 = 6

Answer 6 in

Draw or show your work

56) Challenge: Wait times 2, 3, 3, 4 and 38 minutes. Mean minus median?

Number sentence 10 − 3 = 7

Answer 7 minutes

Draw or show your work

Check-In: Data Displays

Show what you know. No help this time.

Mastery: 4 of 5 (80%)

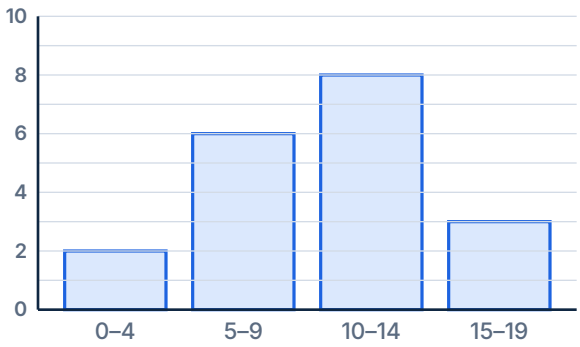
Part 1 Statistics question? Circle yes or no.

1) **How much rain fell each day in May?** (yes) / no

2) **How old is my teacher?** yes / (no)

Part 2 Read the histogram

3) **Laps run**



How many in 15–19? 3

Most in which group? 10–14

How many are 10 or more? 11

How many in all? 19

Part 3 Count the groups

4) **14, 27, 8, 21, 33, 16, 29, 5, 24, 18, 36, 11**

0–9 [2] 10–19 [4] 20–29 [4] 30–39 [2]

Part 4 A far-off number

5) **5, 6, 6, 7, 31: mean = [11] and median = [6]**